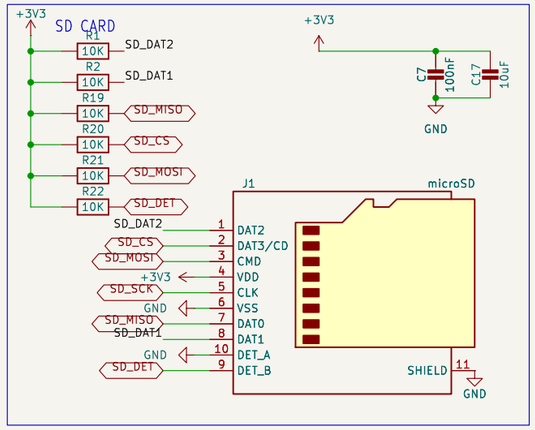
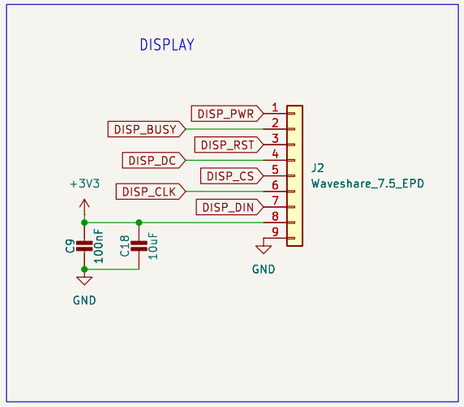
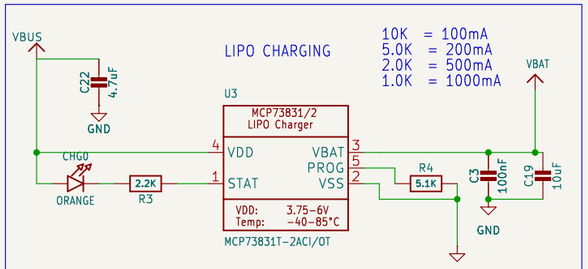
SD CARD



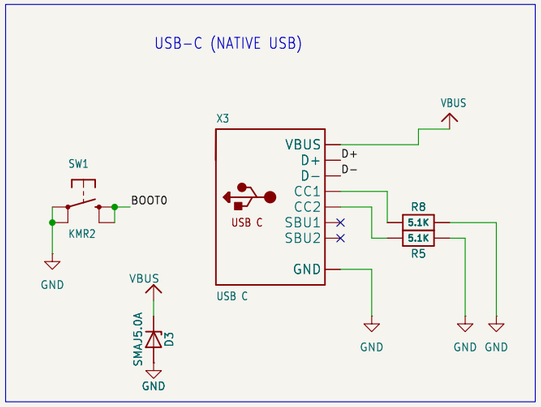
DISPLAY



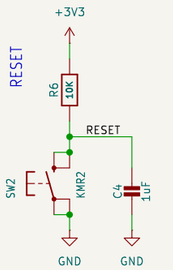
LIPO CHARGING



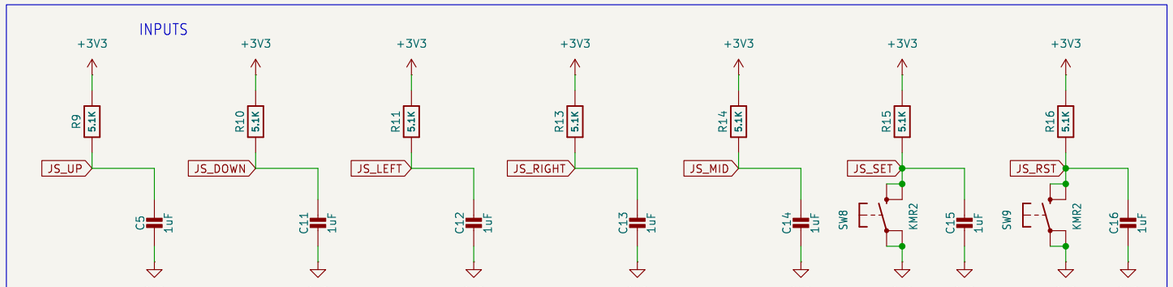
USB-C (NATIVE USB)



RESET

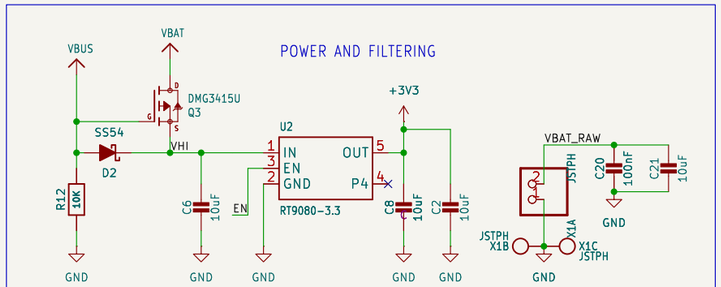


INPUTS

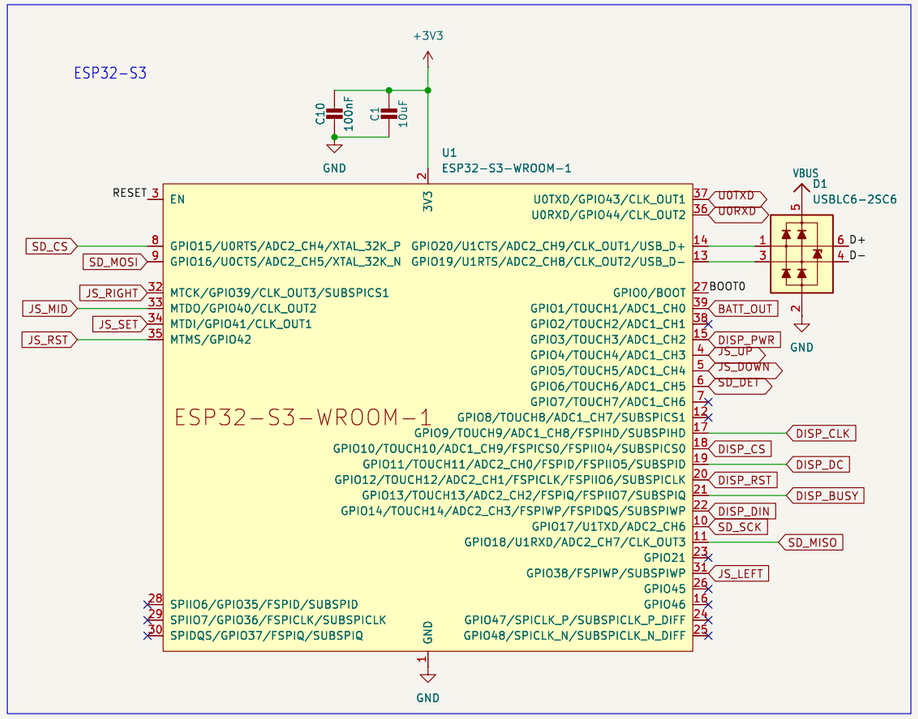


Pull-up + RC debounce per line. JS_UP..JS_MID -> 5-way joystick SW3; SW8 = SET, SW9 = RST

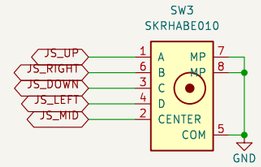
POWER AND FILTERING



ESP32-S3

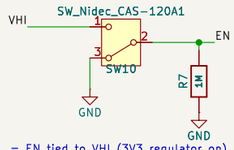


5-WAY JOYSTICK



A/B/C/D = UP/RIGHT/DOWN/LEFT as drawn;
confirm vs. mounting orientation
(remap in firmware if rotated).
Pull-ups + RC debounce: INPUTS block.

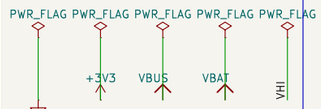
POWER SWITCH



ON = EN tied to VHI (3V3 regulator on)
OFF = EN tied to GND (regulator off,
charging still works on USB)

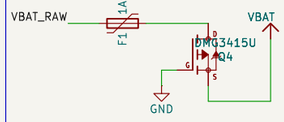
SW10 = SW_SPDT_321: pin 2 (B) is the COMMON.
Pin 1 (C) and pin 3 (A) are the throws.

POWER FLAGS



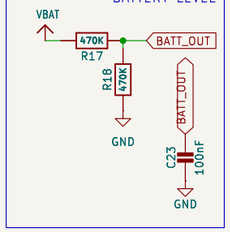
BATTERY PROTECTION

X1: pin 1 = GND (−), pin 2 = BATT+. Check cell leads!

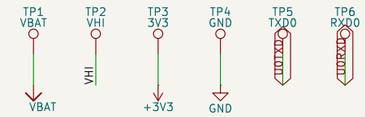


Q4: reverse-polarity protection. Correct cell: body diode conducts, then Vgs = −Vbat turns the channel on (also lets the charger push current back in). Reversed cell: Q4 off.
F1 now protects the whole board from the cell.

BATTERY LEVEL



TEST POINTS



Pads only (not assembled). TXD0/RXD0 = GPIO43/44 UART0 fallback console; native USB-CDC is the primary port.

robertengineer.com

Sheet: /
File: e_reader_v0.1.kicad_sch

Title: EReader

Size: A3	Date: 2026-11-05	**Rev: 0.2.0**
KiCad E.D.A. 9.0.9		Id: 1/1